

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

Yassh Singh

Supervisors: Dr. Anthony Aighobahi, Dr. Mridula Sharma
Thompson Rivers University — UREAP 2026

September 2026

Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it (its falsifier), every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [60]. HVAC accounts for up to half of an individual building’s consumption [7] and for about 30% of all commercial building energy in the United States [20], and 15–30% of that is wasted through faulty operation and poor control [7, 34]; correcting the common faults saves 5–30% of whole-building energy [7, 55]. The faults persist even where detection is installed. Across

60,000 pieces of equipment in 317 commercial buildings, Crowe and colleagues found 245 faults per building per month and 40% of air handlers in a faulted state on any given day, in buildings that run modern automation systems with alarms [12]. Seven of their ten most common faults are behaviour-based: wrong schedules, unreleased overrides, missed setpoint resets, broken economiser sequences. A threshold alarm or a pre-written rule [57] catches such a fault only if someone wrote that rule in advance [26]. The public benchmarks used in this paper seed component, sensor and control-loop faults and none of that class, so this argument motivates the design and is not tested by it.

Research has gone the other way. Machine learning has been surveyed across 211 HVAC fault-detection studies [7], with accuracies saturated at 97–99% on the standard benchmarks [54, 59] and interpretability added afterwards at the level of input features [11, 15, 38, 68, 72], which tells an operator that a sensor mattered, not what the building should have been doing. Of 65 primary studies audited in 2025, two shared code links, one of them broken, and 72% do not say whether their data is public [45].

What an operator needs is a detector that works from the trend data already recorded, names the unit and subsystem, explains the alarm in the physics of the equipment, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge no published work applies event-log process mining to building HVAC fault detection. The 211-study survey contains no process-mining entry [7], the reviews of process mining in Industry 4.0 and on sensor data contain no building HVAC application [2, 8], the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [25], and our own exact-phrase searches of the OpenAlex corpus return none [53].

STRATA applies process mining in stratified form: sensor streams are abstracted into events along ASHRAE Guideline 36 control sequences [3], Petri nets are discovered per device and per unit [46, 61], and the channels that read the log sit beside physics rules and residuals that read the raw signals, each calibrated on its own noise. Our first hypothesis, that the discovered net would itself be the detector, was pre-registered and refuted: a single hand-written rule out-covered the model channel where it was strongest. What settled is the division of labour, and it is a productive one. The strata and the event abstraction give every alarm its location, the frequency and oscillation channels are the only ones that catch the unstable-controller faults, invariants that discovery surfaced became several of the rules that detect, and conformance checking of the discovered nets, the part we expected to detect, contributes no detection of its own. Process mining helps to get detection but is not detecting on its own.

The paper contributes (1) the framework and the detector it yields, evaluated on five public benchmark systems including the two fan-powered terminal units for which no detection results had been published before; (2) what process mining contributes to HVAC fault detection, established under pre-registration: the first such application, whose structure gives the detector its location and several of its rules, and whose conformance channel catches no fault the calibrated channels miss; (3) an evaluation protocol of pre-registration, honoured falsifiers, independent recomputation and full regeneration of every committed result file (artefact); and (4) eight machine-verified errata in the benchmark datasets.

2 Research Aim and Objectives

This research set out to build a fault detector for building HVAC systems that an operator can trust and act on: one that works from the trend data the building already records, explains each alarm in the physics of the equipment, names the unit and the subsystem at fault, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining was the means chosen to get there, because it is the one methodology built to model the sequence in which a system moves between states, and no published work had applied it to this problem. The aim therefore carried a second question: what the discovered process models contribute to such a detector, and whether they detect on their own. Everything reported here is validated on public benchmark data; deployment on an operating building is the intended next phase, and Section ?? states what it would take.

A building operator today finds a fault by waiting for a complaint or by watching trend graphs for a curve that has drifted from where it usually sits; both find it late and neither names the equipment responsible. STRATA is meant to consume a building’s ordinary one-minute trend data and one configuration file naming its sensors and control rules, learn from a fault-free period what healthy operation looks like, and then, every day, name the days that break a physical relationship the healthy period established: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The output is a work order with a false-alarm budget the operator can hold the system to, and onboarding a new equipment type is a configuration exercise. That intent fixes six objectives, each reported with what was done and where it stands.

1. Abstract events from sensor data along Guideline 36 control sequences, with fault-signature events held out of model discovery. Each day is a case and every event is emitted by a configured rule: three kinds produce the state events model discovery may see, five produce fault-signature events it may not. The same code, with no building-specific additions, ran on seven LBNL datasets; the alphabet wall is enforced by unit tests and the circularity it prevents was measured. Met.
2. Discover models and establish what they contribute. Nets discovered at device, unit and cross-system strata on every system; their contribution measured under pre-registration on 199 scenarios across six systems. Met: the structure, the event log and the invariants they surfaced are what they contribute; the hypothesis that they would also detect by conformance alone is refuted.
3. Detect against a measured noise floor. Eight channels, every threshold from the fault-free year, every false-alarm rate out-of-sample; the budget held under injected field conditions and on one measured building. Met; the statistical repair that made this true cost seven detections, and that is stated.
4. Benchmark against known answers. 158 of 175 seeded scenarios detected at 26 false-alarm days in 480; against an audited PCA baseline, 64 of 73 to its 61; against the open-fdd Guideline 36 battery held to the same gate, within one detection on the air handler and three times the detections on the terminal units. All three parts executed; the improvement claim over current practice is withdrawn on the air handler.
5. Make every claim checkable. More than thirty pre-registered experiments, ten adverse results and four later fired falsifiers published, every headline number regenerated under a regeneration gate, a test suite that fails if a caveat is dropped. Met.

6. Release the work openly. Repository, configurations, artefacts, reproduction path and eight machine-verified dataset errata are public. Met.

What the objectives add up to is a detector that does what the intent asks on the benchmarks it was tested on, a measured and stated answer to the process-mining question, and a protocol that made both checkable. What they do not yet add up to is a field result; Section ?? states what that would take. A hybrid with machine-learning classifiers was deferred until the models’ own detection had been established, and is future work.

3 Related Work

Machine learning for HVAC fault detection. Bi and colleagues survey 211 AI-based HVAC fault-detection studies from 2013 to 2023 [7]. Accuracies on the standard benchmarks have saturated: 98% across thirteen fault types with a random-forest and support-vector pipeline on RP-1312 data [59], 97% on three real buildings with a convolutional-and-LSTM hybrid whose authors name interpretability as the problem they did not solve [54]. The interpretable line pairs a deep model with a post-hoc explanation: Deng and colleagues’ graph network on the LBNL single-duct data reaches 88% at 500 labelled samples with per-sensor importance as the explanation [15]; attention and relevance propagation do the same on chiller data [11,38,68,72]. The explanation is at the level of input features, not of the control sequence the building was supposed to follow. The opportunity is a method whose model is its explanation.

Unsupervised methods and cross-building transfer. Real buildings rarely carry fault labels, so the strongest recent work is self-supervised: a masked-reconstruction Transformer on an unlabelled heat-pump building [1], and El Mokhtari and McArthur’s autoencoders across five fan coil units at Toronto Metropolitan University, the closest precedent for a Canadian university deployment [16]. Each yields a scalar anomaly score. Cross-building transfer in this literature means transferring learned weights by domain adaptation [17,19,37,75]; what STRATA transfers is a configuration file naming sensors and rules, a different operation, and the discovered models themselves did not transfer (Section ??).

Process mining and the empty quadrant. Process mining discovers models from event logs, checks conformance against them and enhances them with further perspectives [61,62]. Conformance was used directly as a trace-level anomaly detector [6,56] before neural sequence models overtook it on standard log corpora [48]; our own ablation, which finds no fault detected only by conformance, is consistent with that motivation. In adjacent cyber-physical domains the methodology has succeeded: robotic-arm fault diagnosis at an F1 of 98.9% [66], with alignment features feeding downstream classifiers for gains of 30 to 70 F1 points [67]; industrial IoT sensor validation [29]; industrial control systems [47,58]; and pasteurisation, where continuous sensor signals are abstracted into activity sequences and checked against a process specification [44]. The common barrier is event abstraction [9,13,41]. Building HVAC is an empty quadrant: the 211-study survey has no process-mining entry [7], the 47-study Industry 4.0 review [2] and the 36-study sensor-data review [8] contain no building HVAC application, construction-management process mining treats projects as cases and human tasks as activities [42], LBNL’s fault-detection taxonomy has no such leaf [25], and exact-phrase searches of OpenAlex [53] and the ICPM 2025 programme return none.

Rule-based practice. Fault detection in real buildings is rule-based: commercial platforms such as Automated Logic WebCTRL, installed at the Campus Activity Centre of Thompson Rivers University, ship libraries of over one hundred named faults, and Granderson and colleagues’ survey of fourteen commercial tools confirms the style [25, 52]. LBNL’s automated fault-correction line applies hand-coded correction algorithms through BACnet writeback [39, 40, 55]. Both assume the faults of interest can be enumerated in advance. STRATA is complementary: it learns where the building’s own healthy operation is invariant and holds those invariants at a measured false-alarm budget, and its located, evidenced alarms could in principle feed the correction algorithms. Section ?? measures both the strength and the limit of the rule library on the same benchmarks. Hand-designed Petri nets for HVAC [14, 21, 65, 73] are a different use of the formalism: authored by an engineer, with no event log and no discovery, and should not be conflated with the discovered models used here.

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of ordinary building-management-system trend data and a single configuration file naming the building’s sensors and control rules. The configuration file is the only site-specific artefact; no code change was needed to onboard any system reported here. The framework is stratified in two senses: events are abstracted from the control sequence at device and unit granularity, so every alarm is born with a location, and the channels that read the resulting log (absence, frequency and the two conformance channels) sit beside physics rules, residuals and an oscillation channel that read the raw signals, each calibrated on its own noise.

Event abstraction and the alphabet wall. Continuous sensor streams become a discrete event log through control-sequence rules grounded in ASHRAE Guideline 36 [3], following Janssen and colleagues’ abstraction methodology [31] and the event-abstraction taxonomy of van Zelst and colleagues [64]. Each day is a case. Every event is emitted by a configured rule of one of eight kinds: occupancy, operating mode and time window produce state events, the only events admitted to model discovery; mismatch, leak, setpoint deviation, paired residual and envelope residual produce fault-signature events, which already encode engineering suspicion and are withheld from discovery. The partition, the alphabet wall, exists because of circularity, the process-mining form of data leakage [33, 70]: a model discovered over an alphabet that names faults will later “detect” them by recognising the project’s own rules. Measured on the single-duct system against the pre-partition prototype, the model channel’s apparent recall rises from 1.8% to 5.2% without the wall, while the deployed detector’s output is unchanged, because every leaked model detection was already a rules detection. The separation is enforced by unit tests: injecting signature events into a log cannot change a day’s fitness.

Discovery at three strata. Petri nets are discovered from the state-event log with the inductive miner [35, 36] in pm4py [5] at three granularities: the device stratum models an actuator’s lifecycle, the unit stratum the air handler’s daily cycle, and a cross-system grammar supports the transfer analysis of Section ?. Whether a net encodes order at all is checked directly by a reversal probe: every trace is reversed and re-scored. The single-duct net’s fitness falls from 0.886 to 0.553 under reversal; the two fan-powered nets do not move (0.953 and 0.902 to 0.903), so their fitness carries no ordering information.

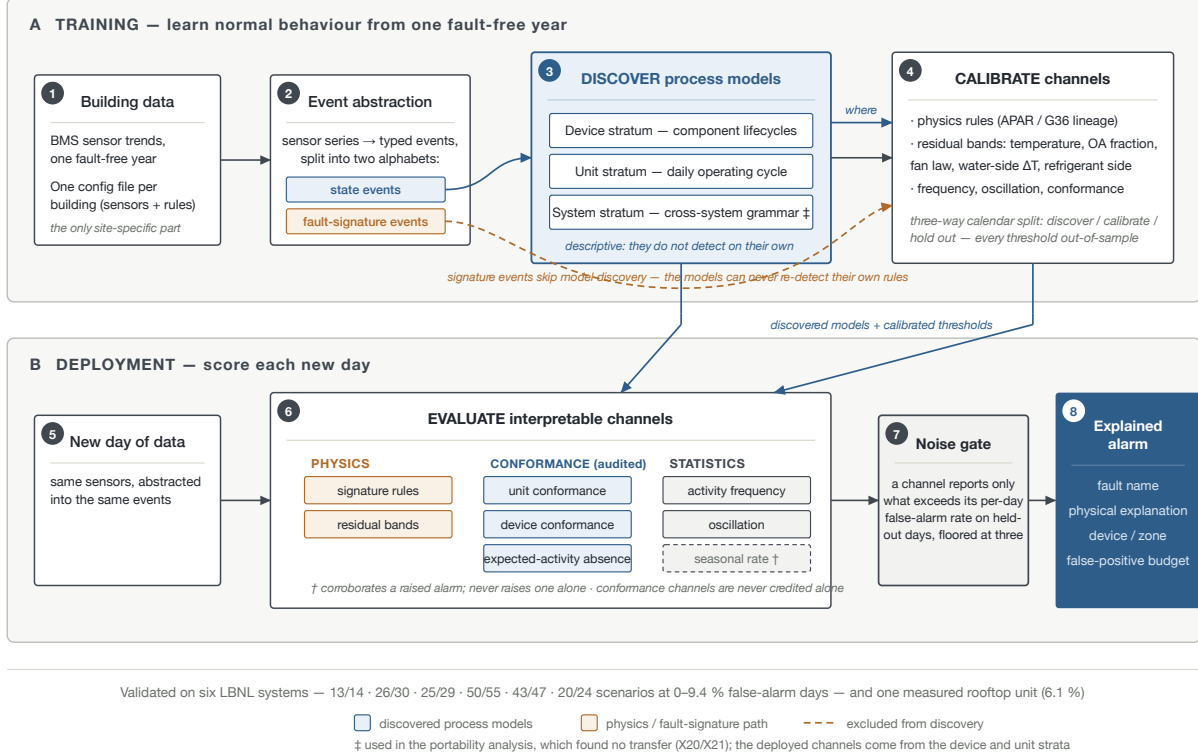


Figure 1: The framework as evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only and supply the strata and the invariants the channels use, but do not detect on their own; every threshold comes from a three-way calendar split, so every false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by the physics, statistical and conformance channels; a channel reports only what exceeds its own measured noise floor; what survives reaches the operator with a name, a physical explanation, the device and the stated budget.

Calibrated channels and the noise gate. Eight channels score each day independently (Table 1): rules from the APAR and Guideline 36 lineage [57] that name direct contradictions between measurements; residual bands on physical relationships that must hold in a known mode; model and device conformance, the alignment cost [10, 63] against the unit and device nets; absence of a scheduled activity; frequency and oscillation bands on activity counts and direction reversals; and a seasonal rate channel that corroborates but may not alarm alone. Calibration uses the fault-free year only, split three ways by calendar: the last eight days of each month (96 days) are held out and report every false-alarm rate; the eight days before them form a calibration slice for the two conformance thresholds; the band channels learn their extremes on every other day and discovery uses the rest. No threshold sees fault data or the days its false-alarm rate is reported on. A channel then reports only what exceeds its own noise floor, a binomial test at $p < 10^{-3}$ against that channel’s holdout false-alarm rate floored at three days [27]; the rules channel, silent on the whole fault-free year by construction, is held to the same rule of three on the year. Detection thresholds are floored at sensor physics (0.5 °F for temperature residuals), which retracted an early result of 21 to 117 detection-days per scenario to at most two. What survives the gate reaches the operator with the implicated device or zone, the evidence, and the channel’s false-alarm budget.

Table 1: The eight channels as deployed. Every threshold is set from the fault-free year alone. The rate channel is advisory: reported, never counted in the detector’s alarm.

Channel	Reads	A day is flagged when	Threshold source
Rules	signature	a control-sequence rule fires during operation	documented bands; rules not silent on the fault-free year are removed at onboarding; gate 3 of 365
Residual	raw signals	a per-mode physical residual leaves its healthy band	[min, max] of training days, widened to a per-rule sensor floor
Model conformance	state	the day’s trace aligns to the unit net below threshold	1 % quantile of fitness on the calibration slice
Device conformance	state	an actuator’s lifecycle trace aligns to its device net below threshold	as above, per device net
Absence	state	a scheduled device produces no lifecycle event on a scheduled day	each device’s healthy silence rate
Frequency	state	a daily activity count leaves its healthy range	[min, max] over training days
Oscillation	raw signals	an actuator’s daily reversal count leaves its healthy range	[min, max] over training days
Rate (advisory)	state	a 30-day active-day rate leaves the range of month-matched healthy windows	training-year windows

5 The Falsification Protocol

An audit of 65 primary machine-learning HVAC fault-detection studies found two that shared code links, one of them broken, and 72% that did not say whether their data was public [45]. A field in that condition needs results that can be checked more than another accuracy figure, so verification was treated as a deliverable. The protocol has three stages and one ledger.

Before a result. Every experiment is pre-registered: the hypothesis and the observation that would refute it are committed to version control before the run [28,49–51]. A falsifier that fires is a finding and is published. Ten results went against the project (four refuted hypotheses, one fault-informed rule, three withdrawn claims, one test without attainable power, one transfer test whose pass a discovery-free control reproduced) and four later tests fired their falsifiers; Appendix ?? lists every experiment with its prediction and outcome. Before any result is computed, a battery of checks runs on the raw data: file hashes, monotonic timestamps, calendar consistency and rotation, Brick coverage [4], and event-log hashes against duplicate scenarios; a healthy-silence gate confirms per configuration that the rules are silent on the fault-free year. Three of the eight dataset errata were found by the gates, the rest by audit, and every one would have silently altered a published result.

Before a number is quoted. An adversarial audit recomputes the claimed quantity independently from raw data, by an auditor briefed to find it wrong (here, AI coding agents, denied the answer key for the sealed test). It caught a baseline detecting file provenance rather than faults; one single-duct detection that was a dataset artefact, so the score is 14 of 14 naive and 13 of 14 adjudicated; a significance test with an impossible null, retracted; detections below sensor precision; and, twice, a completed comparison against commercial practice, the first execution invalid for three reasons and the repaired one for six adapter defects and one defect of design (Section ??). None would have been caught by a test suite, because the code was doing what it was written to do.

After a result. About two hundred tests run on every commit; a regeneration gate re-derives every artefact and diffs it against the stored copy, so a changed value is a defect or a documented decision, never a silent update; every validated dataset is frozen as a fixture. The suite pins published values, fired falsifiers and disclosure obligations: a regeneration that drops a caveat fails the build exactly as a lost feature would. Continuous integration has guarded that a result regenerates [32]; we have not found it reported that it guards that a caveat survives.

What it taught. Sixty-seven numbered lessons carry three patterns: a lesson is not closed until a test enforces it, because documentation decays; every new channel recreated the same missing noise gate until the pipeline enforced it structurally; and simulation flatters, which is why the sensor-precision floor and the noise-floor requirement exist and why the one measured building is reported as a false-alarm rate, not a detection result. The datasets are public by DOI [22–24], the environment is pinned, seeds are fixed, and a claim-to-artefact table maps every published claim to the artefact that backs it.

6 Results

STRATA detects 158 of 175 seeded scenarios on the five simulated systems (14 of 14 naive on the single-duct unit, 13 adjudicated, so 157 under the adjudication) at 26 false-alarm days in 480 held-out fault-free days, at a median of one day, and names the implicated terminal unit in 37 of the 50 attributable detections with none wrong (Table 2). Evaluation began with three public LBNL datasets [23,24]: a single-duct air handler and the parallel and series fan-powered units, the latter two with no previously published detection result; the dual-duct air handler, the fan coil unit and two rooftop units were onboarded afterwards. Scoring follows the day-level, scenario-as-unit conventions of the FDD evaluation literature [18,74]; every number is scored by the detector’s final calibration, the per-day noise floor of Section 4.

System (LBNL)	points	rules (state / sig.)	scored	detected	false alarms	healthy days
SDAHU, single-duct AHU	26	3 / 12	14	14 (13 adj.)	1 / 96	365
PFPU, parallel fan-powered box	109	13 / 45	30	26	8 / 96	365
SFPU, series fan-powered box	109	13 / 49	29	25	9 / 96	365
DDAHU, dual-duct AHU	114	15 / 56	55	50	3 / 96	365
FCU, fan coil unit	29	7 / 19	47	43	5 / 96	365
RTU, simulated rooftop unit	23	3 / 5	24	20	0 / 28	100
RTU, measured rooftop unit (Site 2)	26	2 / 7	1 case	—	3 / 49	182

Table 2: The seven LBNL datasets under the final gate: points mapped, rules declared, scenarios scored and detected, and the out-of-sample false-alarm count on the held-out fault-free days (the last eight of each month). The measured unit carries one fault case and is used for its false-alarm rate only. With every channel counted, including the advisory rate channel, the single-duct rate would be 15.6%.

Time to detection. The median is one day on all six scored systems; the tails are fouling and charge faults: on the dual-duct unit the moderate and severe waterside cooling fouling at 114 days each, on the fan coil unit the moderate heating waterside fouling at 15 days, on the simulated rooftop unit the 20% refrigerant undercharge at 14 days. Gradually developing faults are detected slowly, the expected cost of calibrating below a noise floor. Pooled recall misleads for a gated detector: on the single-duct coil-bias scenarios the residual is physics-bound only inside a detection window (coil off, zone occupied, sustained 120 minutes) that exists on 137 to 164 days of the year, and within those windows recall is 100% on every scenario; the apparent 39–47% ceiling is window prevalence, not capability.

Against a principal-component baseline. A PCA baseline on squared prediction error [30, 69], fitted to 95% of the fault-free variance with its threshold at the largest training-day score and the provenance-leaking columns excluded, detects 61 of the 73 scenarios it shares with STRATA against STRATA’s 64 (13 / 23 / 25 against 13 / 26 / 25), with fewer false alarms on the two terminal units (1 of 96 each against 8 and 9) and within one day on the air handler. Wilson intervals overlap (64 of 73 is [78.2, 93.4]%, 61 of 73 is [73.4, 90.3]%) and an exact McNemar test on the three discordant scenarios gives $p = 0.25$, an absence of power rather than evidence of superiority [43, 71]. The three scenarios only STRATA detects are an unstable VAV damper and the parallel unit’s moderate and severe airside reheat fouling; nine are missed by both, eight of them reheat-coil fouling. What the baseline does not do is name the implicated device or state the evidence behind a flag, which is the difference that matters operationally (Figure 2).

```
STRATA alarm -- series fan-powered unit, 2018-01-01 (day 1 of the fault)
what    zone primary-air flow not tracking its setpoint [channel: rules]
where   terminal unit TU_S, zone S
evidence measured 715 CFM against a 200 CFM setpoint -- a 515 CFM excess
        sustained for 720 min while occupied; damper position 0.80,
        fault-free same-day value 0.32
budget  rules channel: 0 false-alarm days on the 96-day fault-free holdout
seeded  zone-S damper stuck at 80% -- location correct, symptom correct,
        detected on day 1 of 365
```

Figure 2: A worked alarm, scenario SFPU_VAVDMPRStuck_80%, every value read from the committed artefacts and the raw data. The alarm names the symptom and its location, not the root cause.

What the discovered models contribute. For each channel a hand-written equivalent was authored and scored. On the unstable-room-loop scenario two such rules reproduce the frequency channel exactly (141 of 141 flagged days on the series unit, 206 of 206 on the parallel unit): the stratified log’s own count bands make explicit rules an expert could have written, had they known where to look. Transplanted without that knowledge the same rules are unusable, firing on 124 fault-free days on the parallel unit and none on the series unit. Some faults need the classical instrument: the oscillation channel alone carries the series unit’s unstable-damper scenario, 349 flagged days against at most two from any rule. The decisive ablation reads the channel recorded as carrying each detection from the committed results files (the scorecards): no scenario on any system is detected only by the two conformance channels. They are credited on none of the single-duct or parallel-unit detections and on 7 of 25 series-unit detections, never alone; removing them leaves every scorecard unchanged and lowers three systems’ false-alarm rates (9 to 4, 8 to 6 and 5 to 4 of 96). The discovered models contribute the strata and the invariants that shaped the rules; they do not contribute detections.

Localisation and portability. On the 51 detections on the two fan-powered systems whose alarms name a device, the alarm names the correct terminal unit in 37, none in 13 and a wrong unit in none, with three wrong units available in every case; because every seeded zone fault is in the same zone, this is reported as specificity, not discrimination. Over all 158 detections the rule carrying the detection names the seeded component exactly in 109 (69%), the right subsystem but a neighbouring component in 22 (83% at subsystem level), the wrong subsystem in 20 (12.7%) and no component in 7; the pre-registered bar for a component-level claim (wrong at most 10%) was missed, so the claim is subsystem-level. Fourteen of the twenty wrong cases are on the single-zone fan coil unit and name the component that responds to the fault rather than the one that caused it. A pre-registered diagnosis layer of signed residual patterns, redundant-copy consensus and a nearest-pattern family vote raised the exact count to 112 (70.9%, 85% at subsystem level) at a 10.1% wrong-subsystem rate and named the family in 85 of 114 resolved cases with 44 unresolved; both its bars were missed and it is reported, not adopted. Portability was measured: the second system type was onboarded by configuration alone in about 1.5 hours by commit timestamps, and no pipeline code changed for any system.

What the event log carries. Day-by-day comparison of the parallel unit’s fault-free year against its scenarios on the state alphabet shows that under waterside reheat-coil fouling the state-event log is identical to healthy to the minute; nine files have that property, among them every fouling miss that remains on the two fan-powered units, and for them no method that reads that log can detect, process mining or otherwise, because at the deployed alphabet the fault lives in heat-transfer magnitudes the abstraction discards. Four pre-registered vocabulary experiments (X12–X15) tested a time-perspective model, a fault-informed coil residual and an enriched alphabet: the enriched frequency channel alone stays in budget and adds one airside-fouling detection, a scenario the PCA baseline also reaches; nothing else new was found, and richer vocabulary bought detections only at false alarms the budget forbids.

What the conformance channel can see. A pre-registered positive control injected order violations into each system’s fault-free holdout traces. The fan coil and dual-duct nets see order (a reversed day is flagged on 55 of 72 and 59 of 83 holdout days, AUC 0.99); the single-duct net separates it on average (AUC 0.91) but no in-budget threshold sees it, because healthy days fit worse than a reversed day; the two fan-powered nets do not move under any perturbation. A pre-registered grid

of seven process-mining instruments (inductive miner at two noise settings, token replay, heuristics miner, log skeleton, Declare constraints, temporal profile) detects none of the 33 scenarios the deployed detector then missed, at false-alarm rates of 0 to 9.7%. The conformance null is a property of the state-event log, not of the miner or the measure.

Two systems onboarded blind. The dual-duct air handler (114 points, 55 faults in 15 families, two of them new) was onboarded under a pre-registration written before any data file was opened: 80 mappings, 15 state and 26 signature rules, one healthy-silence iteration, no source change; 45 of 55 detected at 3 false-alarm days in 96, 50 of 55 after the coverage audit. The fan coil unit (29 points, 47 scored faults in 17 families, four of them new) followed under the same form: its first run fired two falsifiers because a scoring rule written for the fan-powered units read this unit’s idle setback as a scheduled mode; a two-script repair was pre-registered before it was made, the four earlier scorecards regenerated byte-identically, and the unit scores 41 of 47 at 4 false-alarm days in 96, 43 of 47 after the coverage audit. Nothing on either unit is detected by conformance alone.

The coverage audit. The misses that remained were read as limits of what the datasets record until a read-only scan showed that for six of fourteen the decisive signal was a column no configuration mapped: the configurations had been written from the control sequences, not from an audit of every recorded column. A pre-registered series audited every column of every system, gave each a physically named reader or a written exclusion, and kept every band from the fault-free file with the state alphabet untouched (Table 3). Sixteen scenarios were recovered and none lost for five false-alarm days over 480: airside reheat fouling through the series fan’s power per cfm delivered (“same power, less air”), room-temperature biases through the return air falling away from the biased zone, deck static and temperature biases through the unbiased copies the mixing boxes record (282 of 282 days at zero false alarms), and the fan coil’s damper leaks through the outdoor-air fraction at a fixed damper position. The seventeen misses that remain are coil fouling whose recorded values sit inside every healthy band on every column. Two disclosures bound the claim: which columns to read was decided after a scan of the fault files, so the readers were fault-informed even though every threshold was blind; and five of the sixteen recoveries sit less than one band width outside the healthy extremes, though all five survive bands ten per cent narrower and four of five survive injected field conditions (Section ??).

System	columns mapped	detected, before → after	false alarms, before → after	lost
SDAHU	12 → 26 of 30	14 → 14 / 14	1 → 1	0
PFPU	56 → 109 of 109	22 → 26 / 30	7 → 8	0
SFPU	56 → 109 of 109	21 → 25 / 29	6 → 9	0
DDAHU	79 → 114 of 114	45 → 50 / 55	3 → 3	0
FCU	29 → 29 of 29	40 → 43 / 47	4 → 5	0
all	232 → 387 of 391	142 → 158 / 175	21 → 26 of 480	0

Table 3: The coverage audit: sixteen scenarios recovered, none lost, five false-alarm days added over 480 held-out days, the model and device rows identical on every system.

References

- [1] Mohammad Amin Fatemi Abdollah, Rossano Scoccia, and Marcello Aprile. Transformer encoder based self-supervised learning for HVAC fault detection with unlabeled data. *Building and Environment*, 258:111568, 2024.

- [2] Katsiaryna Akhramovich, Estefanía Serral, and Carlos Cetina. A systematic literature review on the application of process mining to Industry 4.0. *Knowledge and Information Systems*, 66(5):2699–2746, 2024.
- [3] ASHRAE. ASHRAE Guideline 36-2021: High-performance sequences of operation for HVAC systems. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2021.
- [4] Bharathan Balaji, Arka Bhattacharya, Gabriel Fierro, Jingkun Gao, Joshua Gluck, Dezhi Hong, Aslak Johansen, Jason Koh, Joern Ploennigs, Yuvraj Agarwal, Mario Bergés, David Culler, Rajesh Gupta, Mikkel Baun Kjærgaard, Mani Srivastava, and Kamin Whitehouse. Brick: Towards a unified metadata schema for buildings. In *Proceedings of the 3rd ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys '16)*, pages 41–50. ACM, 2016.
- [5] Alessandro Berti, Sebastiaan van Zelst, and Daniel Schuster. PM4Py: A process mining library for Python. *Software Impacts*, 17:100556, 2023.
- [6] Fábio Bezerra and Jacques Wainer. Algorithms for anomaly detection of traces in logs of process aware information systems. *Information Systems*, 38(1):33–44, 2013.
- [7] Jian Bi, Hua Wang, Enbo Yan, Chuan Wang, Ke Yan, Liangliang Jiang, and Bin Yang. AI in HVAC fault detection and diagnosis: A systematic review. *Energy Reviews*, 3(2):100071, 2024.
- [8] Edyta Brzychczy, Milda Aleknonytė-Resch, Dominik Janssen, and Agnes Koschmider. Process mining on sensor data: a review of related works. *Knowledge and Information Systems*, 67:4915–4948, 2025.
- [9] Edyta Brzychczy, Urszula Jessen, Krzysztof Kluza, Sridhar Sriram, and Manuel Vargas Netelstroth. IoT Miner: Intelligent extraction of event logs from sensor data for process mining. *arXiv preprint arXiv:2509.05769*, 2025. Preprint; verified 2026-09-29.
- [10] Josep Carmona, Boudewijn van Dongen, Andreas Solti, and Matthias Weidlich. *Conformance Checking: Relating Processes and Models*. Springer, 2018.
- [11] Zhe Chen, Fu Xiao, Fangzhou Guo, and Jinyue Yan. Interpretable machine learning for building energy management: A state-of-the-art review. *Advances in Applied Energy*, 9:100123, 2023.
- [12] Eliot Crowe, Yimin Chen, Hayden Reeve, David Yuill, Amir Ebrahimifakhar, Yuxuan Chen, Lucas Troup, Amanda Smith, and Jessica Granderson. Empirical analysis of the prevalence of HVAC faults in commercial buildings. *Science and Technology for the Built Environment*, 29(10):1027–1038, 2023.
- [13] Francesca De Luzi, Francesco Leotta, Andrea Marrella, and Massimo Mecella. On the interplay between business process management and Internet-of-Things: A systematic literature review. *Business & Information Systems Engineering*, 67(2):265–288, 2025.
- [14] Wael Deabes, Kheir Eddine Bouazza, and Wasl Alghthami. Smart fuzzy Petri net-based temperature control framework for reducing building energy consumption. *Sensors*, 23(13):5985, 2023.

- [15] Qiao Deng, Zhiwen Chen, Wanting Zhu, Zefan Li, Yifeng Yuan, and Yalin Wang. Adaptive fusion graph convolutional network based interpretable fault diagnosis method for HVAC systems enhanced by unlabeled data. *Energy and Buildings*, 324:114901, 2024.
- [16] Karim El Mokhtari and J. J. McArthur. Autoencoder-based fault detection using building automation system data. *Advanced Engineering Informatics*, 62:102810, 2024.
- [17] Bowei Feng, Qizhen Zhou, Jianchun Xing, Qiliang Yang, Yin Chen, and Zhongkai Deng. Attention-empowered transfer learning method for HVAC sensor fault diagnosis in dynamic building environments. *Building and Environment*, 250:111148, 2024.
- [18] Stephen Frank, Guanjing Lin, Xin Jin, Rupam Singla, Amanda Farthing, and Jessica Granderson. A performance evaluation framework for building fault detection and diagnosis algorithms. *Energy and Buildings*, 192:84–92, 2019.
- [19] Naghmeh Ghalamsiah, Jin Wen, K. Selcuk Candan, Teresa Wu, Zheng O’Neill, and Asra Aghaei. Unsupervised domain adaptation for HVAC fault diagnosis using contrastive adaptation network. *Energy and Buildings*, 337:115659, 2025.
- [20] William Goetzler, Richard Shandross, Jim Young, Oxana Petritchenko, Decker Ringo, and Sam McClive. Energy savings potential and RD&D opportunities for commercial building HVAC systems. Report DOE/EE-1703, U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Building Technologies Office, December 2017. Prepared by Navigant Consulting, Inc.
- [21] Luís Gomes, Anikó Costa, João Paulo Barros, Rui Pais, Tiago Rodrigues, and Richard Ferreira. Petri net based building automation and monitoring system. In 2007 5th IEEE International Conference on Industrial Informatics (INDIN), pages 57–62, 2007.
- [22] Jessica Granderson, Guanjing Lin, Yimin Chen, Armando Casillas, Piljae Im, and Sungkyun Jung. LBNL fault detection and diagnostics datasets. DOE Open Energy Data Initiative (OEDI), dataset, 2022.
- [23] Jessica Granderson, Guanjing Lin, Yimin Chen, Armando Casillas, Jin Wen, Zhelun Chen, Piljae Im, Sen Huang, and Jiazhen Ling. A labeled dataset for building HVAC systems operating in faulted and fault-free states. *Scientific Data*, 10:342, 2023.
- [24] Jessica Granderson, Guanjing Lin, Ari Harding, Piljae Im, and Yan Chen. Building fault detection data to aid diagnostic algorithm creation and performance testing. *Scientific Data*, 7:65, 2020.
- [25] Jessica Granderson, Rupam Singla, Ebony Mayhorn, Paul Ehrlich, Draguna Vrabie, and Stephen Frank. Characterization and survey of automated fault detection and diagnostic tools. Technical Report LBNL-2001075, Lawrence Berkeley National Laboratory, Berkeley, CA, November 2017.
- [26] H. Burak Gunay, Jayson Bursill, Brent Huchuk, and Scott Shillinglaw. Inverse model-based detection of programming logic faults in multiple zone VAV AHU systems. *Building and Environment*, 211:108732, 2022.
- [27] James A. Hanley and Abby Lippman-Hand. If nothing goes wrong, is everything all right? Interpreting zero numerators. *JAMA*, 249(13):1743–1745, 1983.

- [28] Jake M. Hofman, Angelos Chatzimparmpas, Amit Sharma, Duncan J. Watts, and Jessica Hullman. Pre-registration for predictive modeling. *arXiv:2311.18807*, 2023.
- [29] Erik Johannes Husom, Arda Goknil, Felix Mannhardt, Simeon Tverdal, Sagar Sen, and Phu Nguyen. Unsupervised learning and process analysis for sensor data validation in the IIoT. *ACM Transactions on Internet Technology*, 26(2):1–34, 2026.
- [30] J. Edward Jackson and Govind S. Mudholkar. Control procedures for residuals associated with principal component analysis. *Technometrics*, 21(3):341–349, 1979.
- [31] Dominik Janssen, Felix Mannhardt, Agnes Koschmider, and Sebastiaan J. van Zelst. Process model discovery from sensor event data. In *Process Mining Workshops (ICPM 2020)*, volume 406 of *Lecture Notes in Business Information Processing*, pages 69–81. Springer, 2021.
- [32] Ivo Jimenez, Michael Sevilla, Noah Watkins, Carlos Maltzahn, Jay Lofstead, Kathryn Mohror, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. The Popper convention: Making reproducible systems evaluation practical. In *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, 2017.
- [33] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.
- [34] Srinivas Katipamula and Michael R. Brambley. Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I. *HVAC&R Research*, 11(1):3–25, 2005.
- [35] Sander J. J. Leemans, Dirk Fahland, and Wil M. P. van der Aalst. Discovering block-structured process models from event logs — a constructive approach. In *Application and Theory of Petri Nets and Concurrency (PETRI NETS 2013)*, volume 7927 of *Lecture Notes in Computer Science*, pages 311–329. Springer, 2013.
- [36] Sander J. J. Leemans, Dirk Fahland, and Wil M. P. van der Aalst. Discovering block-structured process models from event logs containing infrequent behaviour. In *Business Process Management Workshops (BPM 2013)*, volume 171 of *Lecture Notes in Business Information Processing*, pages 66–78. Springer, 2014.
- [37] Yutian Lei, Cheng Fan, Haihui He, and Yonghang Xie. Enabling efficient cross-building HVAC fault inferences through novel unsupervised domain adaptation methods. *Building and Environment*, 272:112678, 2025.
- [38] Guannan Li, Luhan Wang, Limei Shen, Liang Chen, Hengda Cheng, Chengliang Xu, and Fan Li. Interpretation of convolutional neural network-based building HVAC fault diagnosis model using improved layer-wise relevance propagation. *Energy and Buildings*, 286:112949, 2023.
- [39] Guanjing Lin, Marco Pritoni, Yimin Chen, and Jessica Granderson. Development and implementation of fault-correction algorithms in fault detection and diagnostics tools. *Energies*, 13(10):2598, 2020.
- [40] Guanjing Lin, Marco Pritoni, Yimin Chen, Raphael Vitti, Christopher Weyandt, and Jessica Granderson. Implementation and test of an automated control hunting fault correction algorithm in a fault detection and diagnostics tool. *Energy and Buildings*, 283:112796, 2023.

- [41] Juergen Mangler, Ronny Seiger, Janik-Vasily Benzin, Joscha Grüger, Yusuf Kirikkayis, Florian Gallik, Lukas Malburg, Matthias Ehrendorfer, Yannis Bertrand, Marco Franceschetti, Barbara Weber, Stefanie Rinderle-Ma, Ralph Bergmann, Estefanía Serral Asensio, and Manfred Reichert. From Internet of Things data to business processes: Challenges and a framework. *arXiv:2405.08528*, 2024.
- [42] Araham Jesus Martinez Lagunas and Mazdak Nik-Bakht. Process mining, modeling, and management in construction: A critical review of three decades of research coupled with a current industry perspective. *Journal of Construction Engineering and Management*, 150(11):04024158, 2024.
- [43] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [44] Azin Moradbeikie, Ana Paula Ayub da Costa Barbon, Iuliana Malina Grigore, Douglas Fernandes Barbin, and Sylvio Barbon Junior. Process mining of sensor data for predictive process monitoring: A HACCP-guided pasteurization study case. *Systems*, 13(11):935, 2025.
- [45] Adil Mukhtar, Michael Hadwiger, Franz Wotawa, and Gerald Schweiger. Reproducibility of machine learning-based fault detection and diagnosis for HVAC systems in buildings: An empirical study. *Energy and AI*, 22:100658, 2025. Also *arXiv:2508.00880*.
- [46] Tadao Murata. Petri nets: Properties, analysis and applications. *Proceedings of the IEEE*, 77(4):541–580, 1989.
- [47] David Myers, Suriadi Suriadi, Kenneth Radke, and Ernest Foo. Anomaly detection for industrial control systems using process mining. *Computers & Security*, 78:103–125, 2018.
- [48] Timo Nolle, Stefan Luetzgen, Alexander Seeliger, and Max Mühlhäuser. BINet: Multi-perspective business process anomaly classification. *Information Systems*, 103:101458, 2022.
- [49] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018.
- [50] Samuel Pawel, Lucas Kook, and Kelly Reeve. Pitfalls and potentials in simulation studies: Questionable research practices in comparative simulation studies allow for spurious claims of superiority of any method. *Biometrical Journal*, 66(1):e2200091, 2024. First published online 2023; also *arXiv:2203.13076* (2022).
- [51] Cameron Pitsch, Tobias Brockhoff, Jan Niklas Adams, Sander J. J. Leemans, Leo Anthony Celi, and Wil M. P. van der Aalst. Hypothesis testing for processes. In *2025 7th International Conference on Process Mining (ICPM)*, pages 1–8. IEEE, 2025.
- [52] Ojas Pradhan, Jin Wen, Yimin Chen, Xing Lu, Mengyuan Chu, Yangyang Fu, Zheng O’Neill, Teresa Wu, and K. Selçuk Candan. Dynamic Bayesian network-based fault diagnosis for ASHRAE Guideline 36. In *Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys ’21)*, pages 365–368. ACM, 2021.
- [53] Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv:2205.01833*, 2022.

- [54] Prince, Byungun Yoon, and Prashant Kumar. Fault detection and diagnosis in air-handling unit (AHU) using improved hybrid 1D convolutional neural network. *Systems*, 13(5):330, 2025.
- [55] Marco Pritoni, Guanqing Lin, Yimin Chen, Raphael Vitti, Christopher Weyandt, and Jessica Granderson. From fault-detection to automated fault correction: A field study. *Building and Environment*, 214:108900, 2022.
- [56] Anne Rozinat and Wil M. P. van der Aalst. Conformance checking of processes based on monitoring real behavior. *Information Systems*, 33(1):64–95, 2008.
- [57] Jeffrey Schein, Steven T. Bushby, Natascha S. Castro, and John M. House. A rule-based fault detection method for air handling units. *Energy and Buildings*, 38(12):1485–1492, 2006.
- [58] Prasannjeet Singh, Mehdi Saman Azari, Francesco Vitale, Francesco Flammini, Nicola Mazzocca, Mauro Caporuscio, and Johan Thornadtsson. Using log analytics and process mining to enable self-healing in the Internet of Things. *Environment Systems and Decisions*, 42:234–250, 2022.
- [59] Wunna Tun, Johnny Kwok-Wai Wong, and Sai-Ho Ling. Hybrid random forest and support vector machine modeling for HVAC fault detection and diagnosis. *Sensors*, 21(24):8163, 2021.
- [60] United Nations Environment Programme and Global Alliance for Buildings and Construction. Global status report for buildings and construction 2024/2025: Not just another brick in the wall. Technical report, United Nations Environment Programme, Nairobi, 2025. Buildings and construction: 32 % of global final energy use and 34 % of energy-related CO₂ emissions in 2023.
- [61] Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2nd edition, 2016.
- [62] Wil M. P. van der Aalst, Arya Adriansyah, Ana Karla Alves de Medeiros, Franco Arcieri, Thomas Baier, Tobias Blickle, Jagadeesh Chandra Bose, Peter van den Brand, Ronald Brandtjen, Joos Buijs, et al. Process mining manifesto. In *Business Process Management Workshops (BPM 2011)*, volume 99 of *Lecture Notes in Business Information Processing*, pages 169–194. Springer, 2012.
- [63] Wil M. P. van der Aalst, Arya Adriansyah, and Boudewijn van Dongen. Replaying history on process models for conformance checking and performance analysis. *WIRES Data Mining and Knowledge Discovery*, 2(2):182–192, 2012.
- [64] Sebastiaan J. van Zelst, Felix Mannhardt, Massimiliano de Leoni, and Agnes Koschmider. Event abstraction in process mining: literature review and taxonomy. *Granular Computing*, 6(3):719–736, 2021.
- [65] Emilia Villani and Paulo E. Miyagi. A hybrid Petri net modeling approach for HVAC systems in intelligent buildings. *SBA: Controle & Automação*, 15(2):135–148, 2004.
- [66] Francesco Vitale, Nicola Dall’Ora, Sebastiano Gaiardelli, Enrico Fraccaroli, Nicola Mazzocca, and Franco Fummi. Process mining-driven modeling and simulation to enhance fault diagnosis in cyber-physical systems. *Journal of Manufacturing Systems*, 84:189–206, 2026. Also arXiv:2506.21502 (2025).

- [67] Francesco Vitale, Marco Pegoraro, Wil M. P. van der Aalst, and Nicola Mazzocca. Control-flow anomaly detection by process mining-based feature extraction and dimensionality reduction. *Knowledge-Based Systems*, 310:112970, 2025.
- [68] Hong Wang, Wenjing Yan, Yajuan Zhang, Jianqiang Yu, Wensheng Gao, Bowen Zhou, and Yi Liu. xMHAC-PDCNN: An explainable deep network with spatial feature embedding for fault detection and diagnosis of chillers under variable operating conditions. *Building and Environment*, 284:113504, 2025.
- [69] Shengwei Wang and Fu Xiao. AHU sensor fault diagnosis using principal component analysis method. *Energy and Buildings*, 36(2):147–160, 2004.
- [70] Hans Weytjens and Jochen De Weerd. Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring. In *Business Process Management Workshops (BPM 2021)*, volume 436 of *Lecture Notes in Business Information Processing*, pages 18–29. Springer, 2022.
- [71] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.
- [72] Chenglong Xiong, Guannan Li, Ying Yan, Hanyuan Zhang, Chengliang Xu, and Liang Chen. Effects of various information scenarios on layer-wise relevance propagation-based interpretable convolutional neural networks for air handling unit fault diagnosis. *Building Simulation*, 17(10):1709–1730, 2024.
- [73] Bing Xu, Jun Su, Qiu Min Fan, and Ya Cheng Zhu. Wireless sensors system for the chillers fault diagnostic based on characteristic Petri net modelling. *International Journal of Wireless and Mobile Computing*, 9(1):43–48, 2015.
- [74] David P. Yuill and James E. Braun. Evaluating the performance of fault detection and diagnostics protocols applied to air-cooled unitary air-conditioning equipment. *HVAC&R Research*, 19(7):882–891, 2013.
- [75] Xu Zhu, Kang Chen, Burkay Anduv, Xinqiao Jin, and Zhimin Du. Transfer learning based methodology for migration and application of fault detection and diagnosis between building chillers for improving energy efficiency. *Building and Environment*, 200:107957, 2021.